

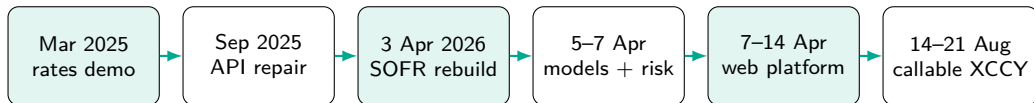
# Building a QuantLib Derivatives Workbench

Human-directed development with Codex and Claude, from swaps to callable XCCY

Tim Launer + two coding agents

Evidence through 22 August 2026

# The product and its collaborative method matured together



- Git records the major releases; local birth times refine the one-commit August implementation sequence.
- Tim directed the economics; Codex built most specifications, code, integrations, and tests; Claude added review, validation, and regression depth.

**Takeaway.** The repository evolved with the human-agent team: every product increment also strengthened the method used to build the next one.

- The product defines dated, signed cash flows and exercise rights; the market supplies curves, volatility, FX, correlations, and fixings.
- The model maps uncertainty into state dynamics; the numerical method converts those dynamics into values and risks.
- Dates, currencies, day counts, collateral, quote conventions, and measures are economic inputs rather than metadata.

**Takeaway.** Sophisticated numerics cannot repair an incorrectly specified contract.

- The March 2025 release connected a simple curve, swap, European swaption, and Bermudan through Python bindings.
- The initial Bermudan incorrectly used a European Black engine and an unsuitable underlying swap definition.
- September repairs introduced a forward-starting underlying and Hull–White TreeSwaptionEngine, while fixing changed QuantLib APIs.

**Takeaway.** Compact examples become reliable only when product, engine, conventions, and smoke tests agree.

## Market construction

Overnight front point plus SOFR OIS  
helpers spanning short tenors to 30 years;  
log-cubic discount interpolation and explicit  
settlement conventions.

## Evidence

Every helper reprices to its quote;  
discount, zero, and one-day forward views  
expose curve shape and interpolation  
behavior.

**Takeaway.** The curve is a calibrated market object with residuals, not merely a line drawn through rates.

- A diagonal strip maps each future exercise date to the tenor of its remaining swap.
- Normal-volatility quotes become QuantLib swaption helpers priced with a Jamshidian engine under Hull–White.
- Mean reversion is governed; a constant volatility minimizes declared helper pricing errors and reports every residual.

**Takeaway.** Calibration is reproducible only when helper selection, units, objective, bounds, and stopping rules are visible.

$$V(t_i, x) = \max(E(t_i, x), \mathbb{E}[D_{i,i+1} V(t_{i+1}, X_{i+1}) \mid X_i = x]).$$

- Hull–White contributes a one-dimensional Gaussian short-rate state and an exact fit to today's curve.
- A recombining lattice propagates continuation values backward across business-day-adjusted exercise dates.
- Tree steps are a numerical parameter and require convergence checks.

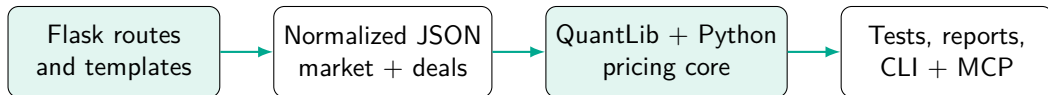
**Takeaway.** The tree is natural for a single-currency, one-factor stopping problem, not a universal callable engine.

- G2++ added a second Gaussian rates model; governed curve, volatility, and parameter bumps exposed calibration dependency.
- An SPX cliquet added Black–Scholes forward-start values, analytic Greeks, Monte Carlo distributions, and path-dependent scheduling.
- Parity, decomposition, scenario, common-random-number, and bump tests checked signs, scaling, and nonlinear response.

**Takeaway.** New products mattered because they expanded the repository's validation vocabulary, not merely its menu.

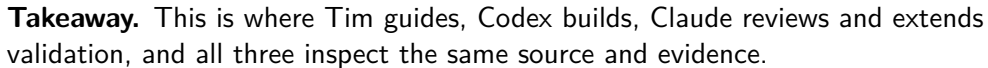


## A small technical stack became the agents' shared workbench

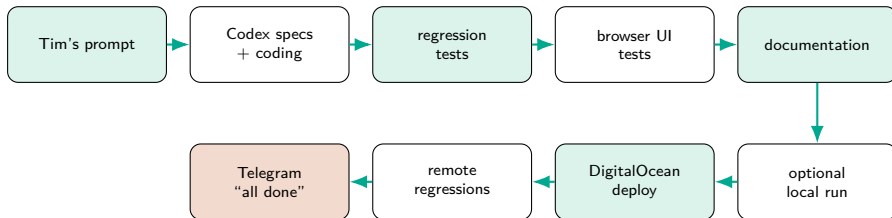


- On the Mac, Git makes Codex and Claude changes, specifications, diffs, tests, JSON evidence, and server logs mutually inspectable.
- QuantLib supplies proven C++ primitives; Python makes the extension transparent; Apache, Gunicorn, and systemd run it on DigitalOcean.

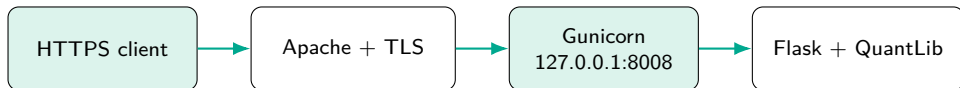
**Takeaway.** One repository connects human direction, two agents, open-source quant infrastructure, local evidence, and hosted execution.



# One prompt can run the complete delivery chain



**Takeaway.** No manual handoff is required after the prompt; Claude's validation extensions and remote regressions protect later Codex changes.



- systemd runs the app under www-data at `/var/www/html/tgir_quantlib_tools`; secrets stay in a mode-600 `.env`.
- Deployment builds a fresh final-path virtualenv, restarts the service, checks loopback and public TLS health, and rolls back on failure.

**Takeaway.** For about USD 7/month, this serious hobby gains a hosted evidence lane with TLS, process, health-check, regression, and rollback controls.

## Native building blocks

Curves, calendars, schedules, XCCY swaps, Hull–White/G2, swaption helpers and engines, stochastic processes, and Gaussian path generators.

## Missing composition

A domestic-measure joint rates/FX process, callable cross-currency cash-flow semantics, and a multi-date optimal-stopping engine.

**Takeaway.** Native support is a spectrum: reuse mature primitives while owning the missing economics.

- The example is a ten-year EUR/USD constant-notional swap, non-callable for two years and then annually cancellable.
- The holder receives quarterly compounded EUR ESTR and pays semiannual USD fixed at 4%.
- USD 100 million and EUR 90.909 million notionals exchange initially and finally; USD is collateral and reporting currency.

**Takeaway.** Exercise means cancelling the remaining tail under explicit event ordering, not receiving a generic option payoff.

$$\begin{aligned}dr_d &= [\theta_d(t) - a_d r_d]dt + \sigma_d dW_d, \\dr_f &= [\theta_f(t) - a_f r_f - \rho_{fS} \sigma_f \sigma_S(t)]dt + \sigma_f dW_f, \\d \log S &= [r_d - r_f - \frac{1}{2} \sigma_S^2(t)]dt + \sigma_S(t) dW_S.\end{aligned}$$

- Dynamics are under the USD money-market measure with  $S = \text{USD per EUR}$ .
- The full  $3 \times 3$  Brownian correlation matrix must be positive semidefinite.

**Takeaway.** FX quotation and numeraire determine both the FX drift and the sign of the foreign quanto correction.

- USD SOFR discounts domestic cash flows; EUR ESTR forecasts coupons; FX forwards imply the USD-collateralized effective EUR discount curve.
- One constant Hull–White sigma per currency fits diagonal normal swaptions with governed mean reversions.
- Piecewise FX sigmas reproduce selected ATM Black maturities after including stochastic-rate contributions to terminal log-FX variance.

**Takeaway.** The exact relative RMS price error and instrument-level residuals disclose the cost of the fixed-volatility approximation.



- The state contains both Hull–White factors,  $\log FX$ , and the integrated domestic short rate.
- Each interval uses exact Ornstein–Uhlenbeck decay and an integrated covariance matrix; eigenfactorization handles near-singular cases.
- The event grid contains coupon, fixing, payment, notional, exercise, and volatility-bucket dates, with intermediate step caps.

**Takeaway.** Exact conditional transitions reduce discretization bias but do not eliminate sampling error or model risk.

- ① Simulate training paths and move backward through the annual call dates.
- ② Regress discounted future cancellable-tail value on standardized rates, log FX, leg values, squares, and cross-products.
- ③ For zero-fee cancellation, exercise when estimated continuation value is negative.
- ④ Freeze the policy and apply it forward on an independently seeded pricing sample.

**Takeaway.** Out-of-sample policy valuation removes in-sample look-ahead optimism and reports an honest lower-bound estimate.

## Research evidence

Stored callable NPV is near USD 8.34 million versus non-callable NPV near USD 0.69 million. Residuals, martingales, exercise profiles, convergence, and bump stability qualify the mark.

## Collaborative engineering path

Codex extends the specified core; Claude's regression layer challenges changes; Python keeps evidence transparent, while profiled C++ can add native QuantLib types and scalable simulation.

**Takeaway.** The system is designed to extend QuantLib rigorously: stabilize economics and tests first, then migrate measured bottlenecks to C++.

The compounding achievement

**Tim directs → Codex specifies and builds**

**→ Claude reviews and extends validation**

**→ Mac proves → DigitalOcean hosts**

**A serious hobby: rigorous quantitative engineering from one prompt to tested deployment and a Telegram “all done”—for about USD 7/month.**